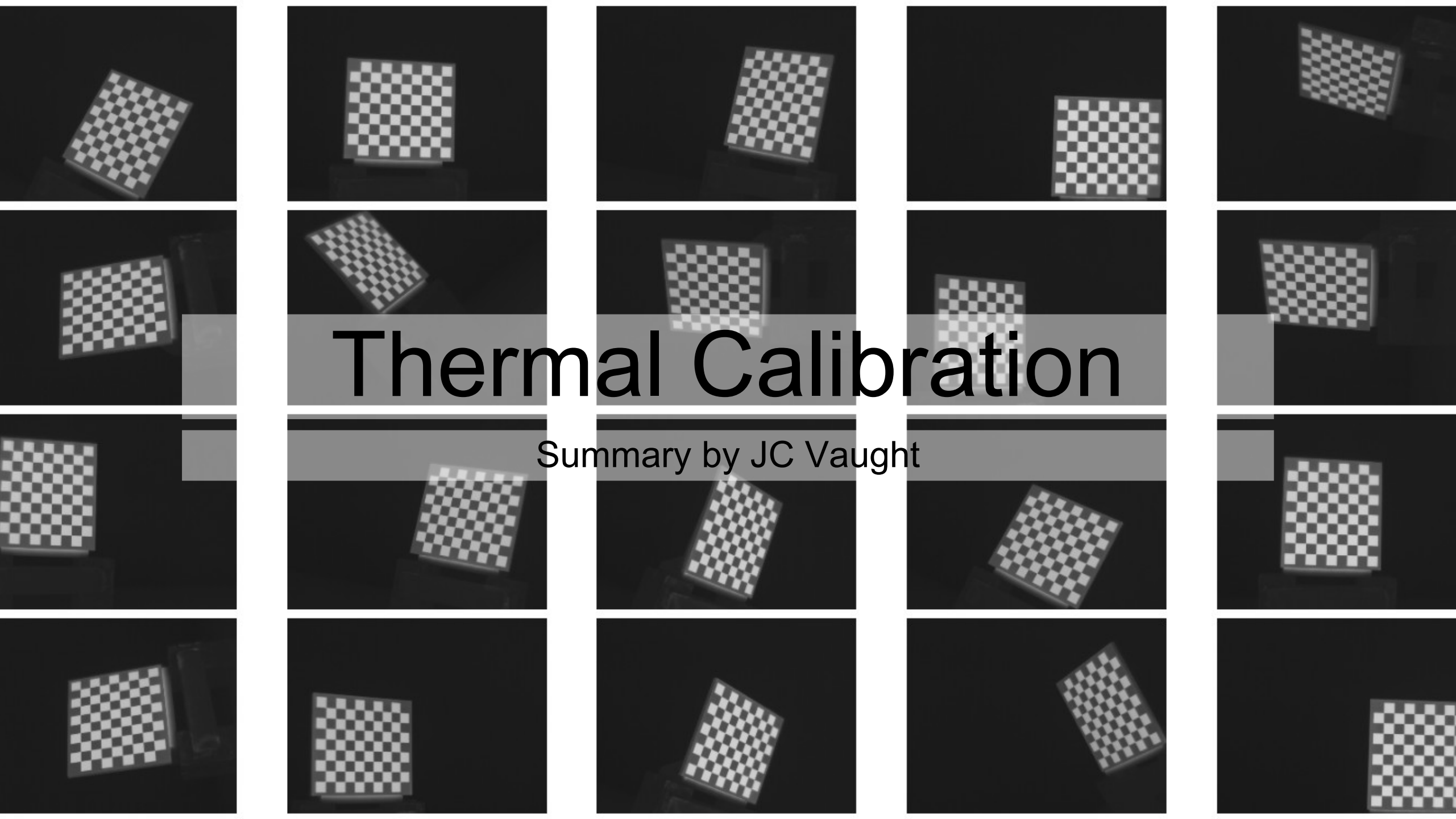




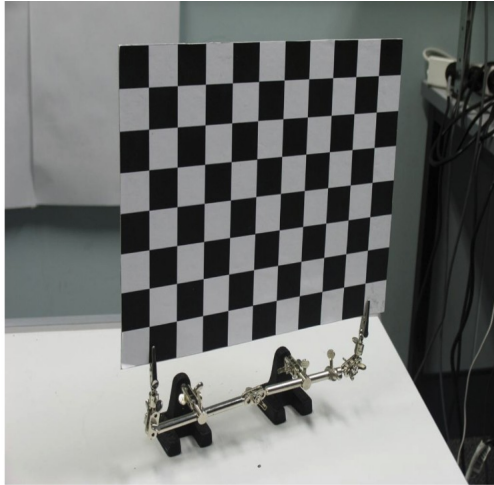
Thermal Calibration

Summary by JC Vaught

Primary categories – according to me, myself, & I

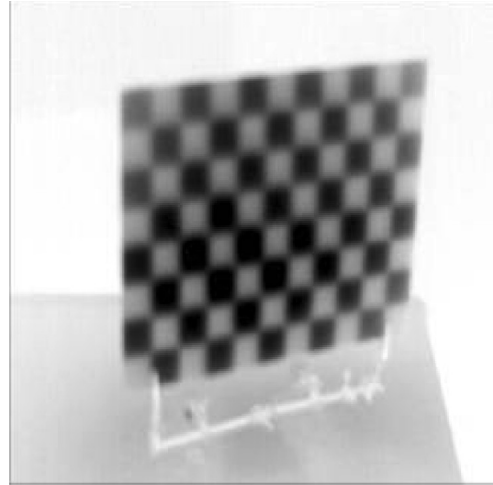
- Active – I define as ‘needing an external source for ΔT ’
 - Lamp/Bulb Array - For example, Christmas lights
 - Heat-Producing Resistors - Can use nails for conducting heat
 - Peltier Modules - For active cooling and heating of specific squares
- Passive – I define as ‘heating once, or using environment for ΔT ’
 - Paper Checkerboards - Heated by a flood lamp.
 - Cardboard Cutouts - A heat source is placed behind the cardboard.
 - Taped Patterns - A reflection-based model (i.e., heat source is sky or env).
 - Paint/Print on Metal - Also a reflection-based model.

Passive – Paper Checkerboards

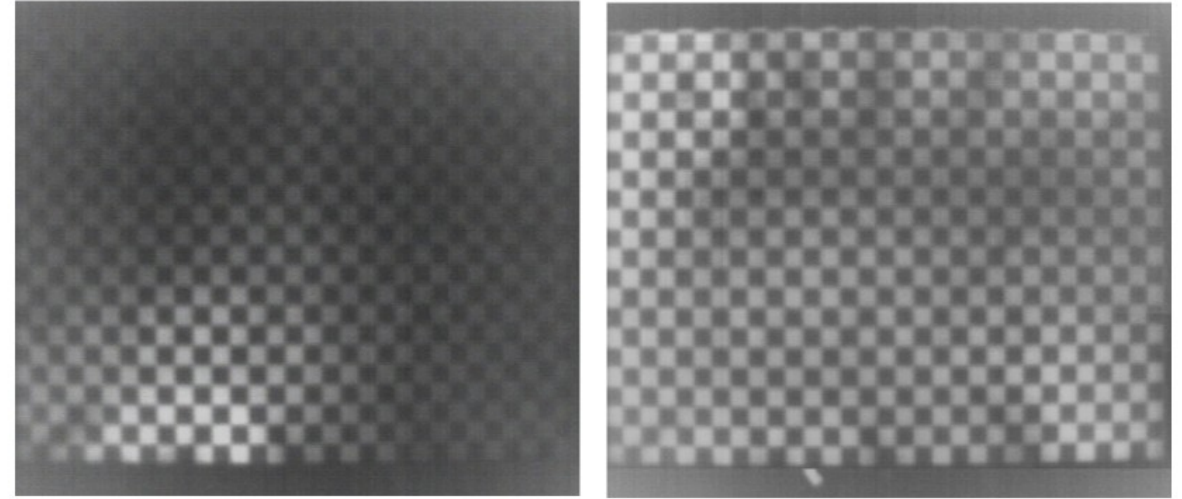


(a) Left camera.

Flood Lamp
RMSE 0.41-0.48

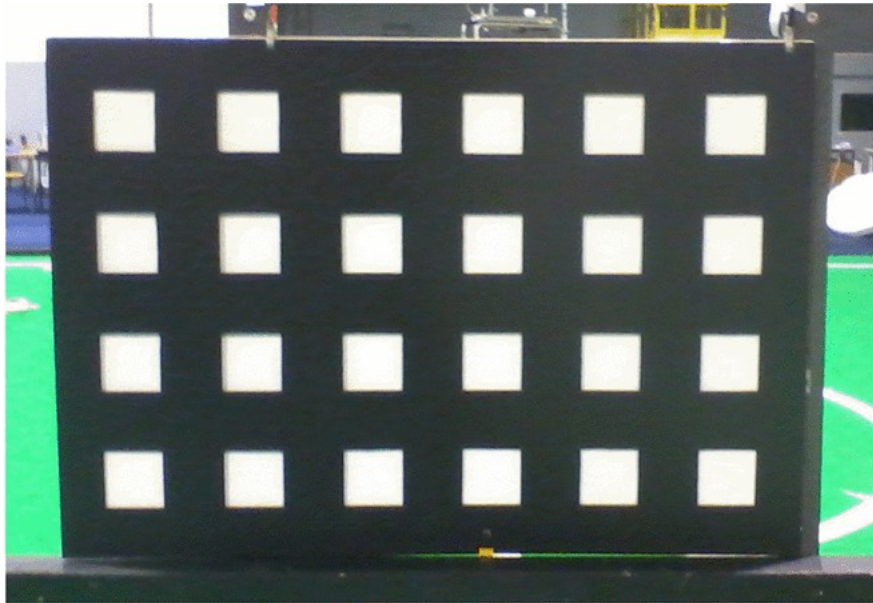


(b) Thermal camera.

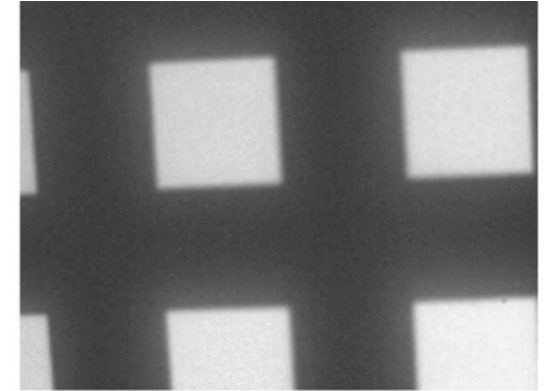
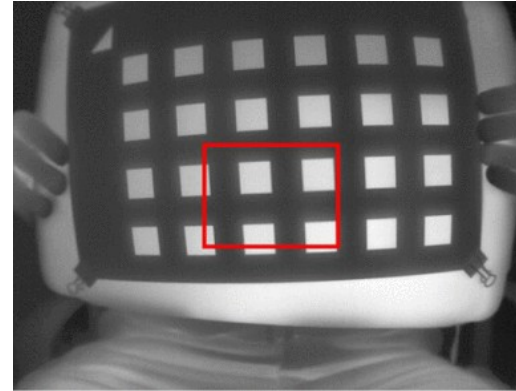


IR Lamp version (also uses frame
combination(i.e. HDR))
RMSE 0.6-0.714

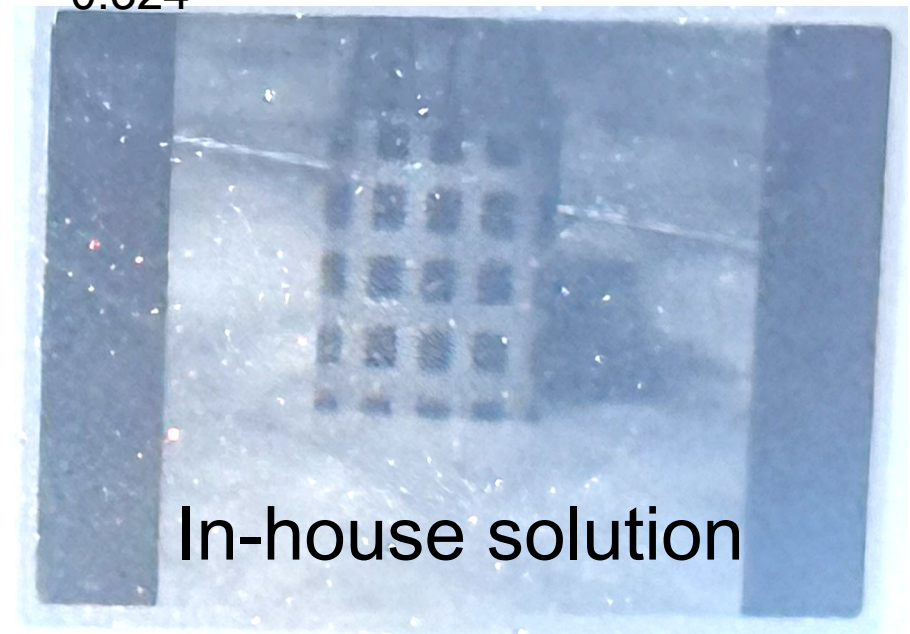
Passive – non-standard mask



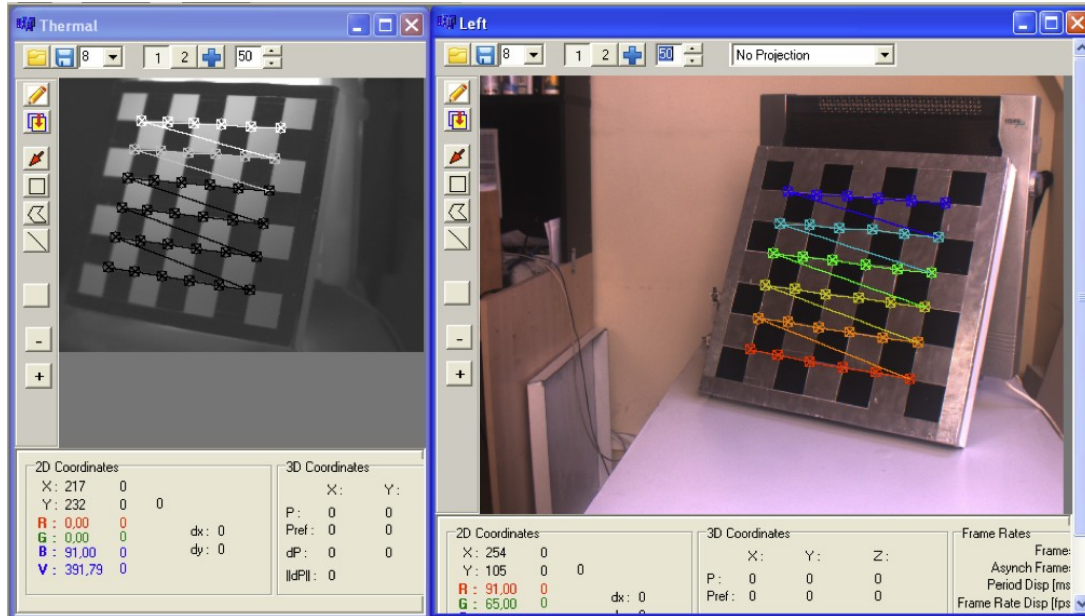
Unknown heat
source



Computer monitor – 0.284
-0.324

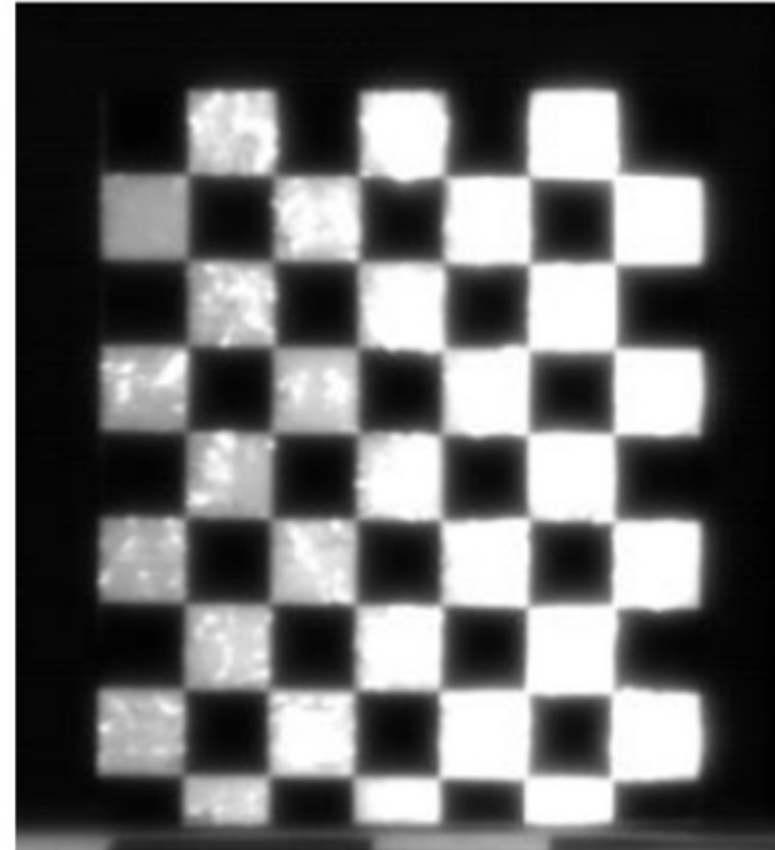


Passive – metal tape on board



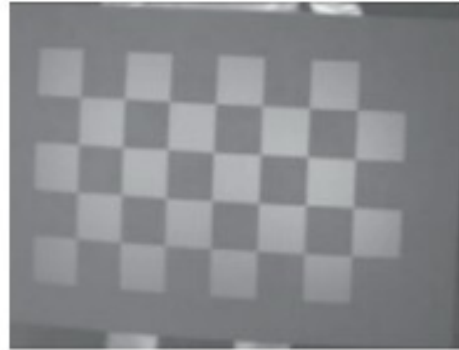
Aluminum foil
Heat = home
radiator

* Note this is very common
method

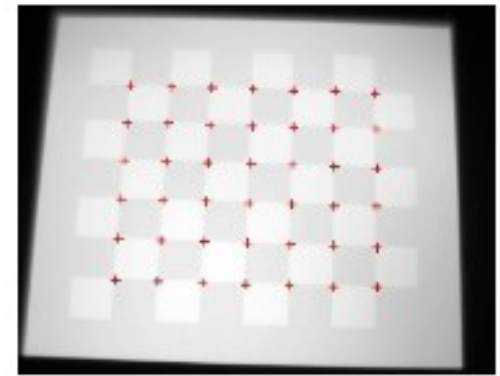
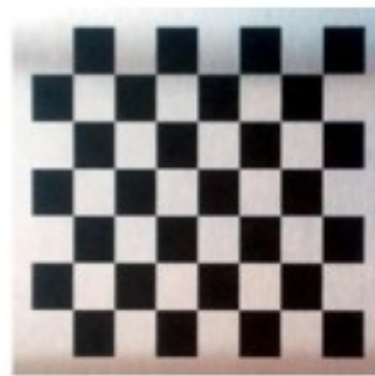


Passive – print on metal

- Aluminum plate; printed pattern
- RMSE 0.10 - 0.22



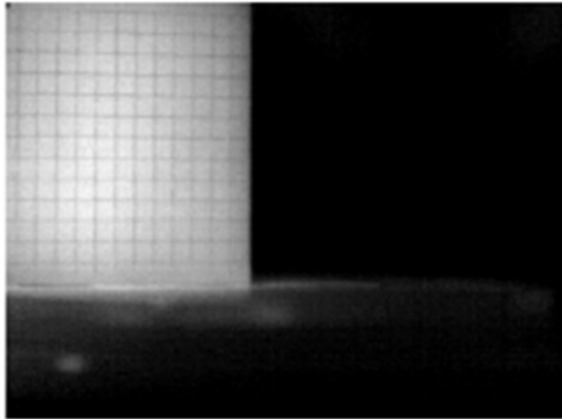
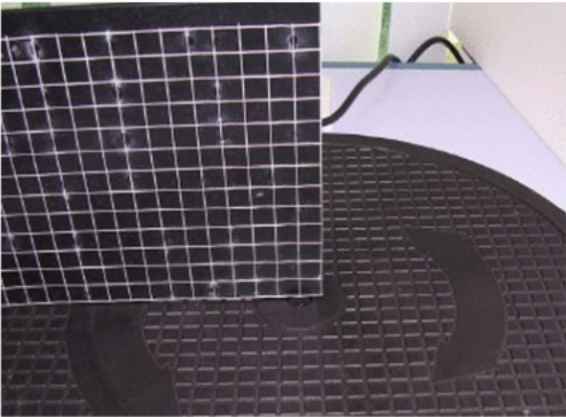
- Aluminum plate; UV printed pattern
- RMSE 0.233



* Note this is very common method

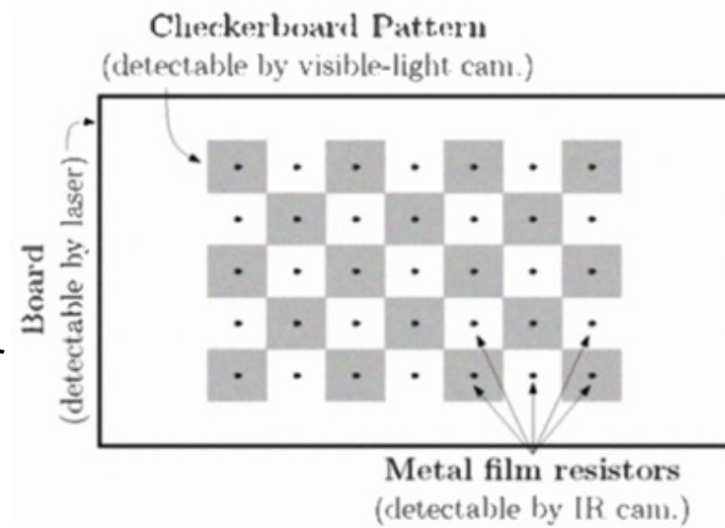
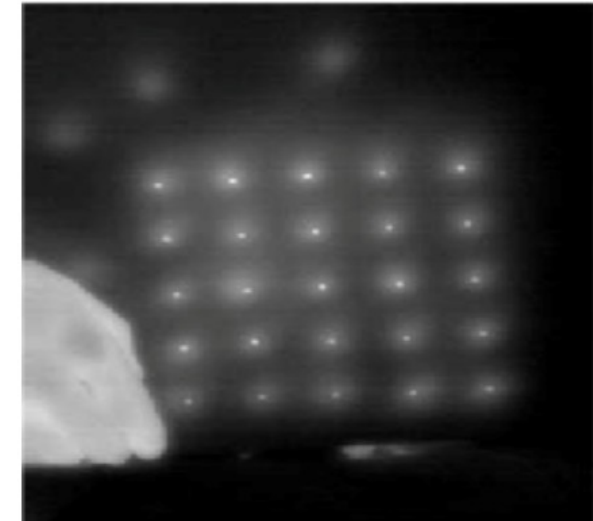
Active – heat points

- Metal wires heated
- RMSE 0.175-0.177



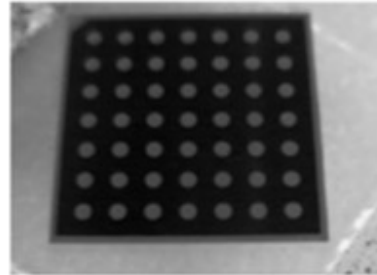
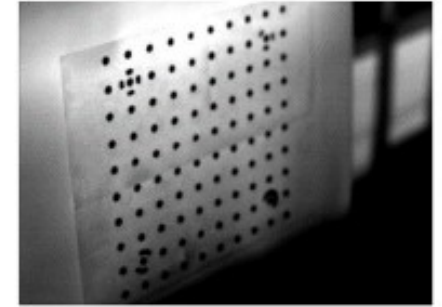
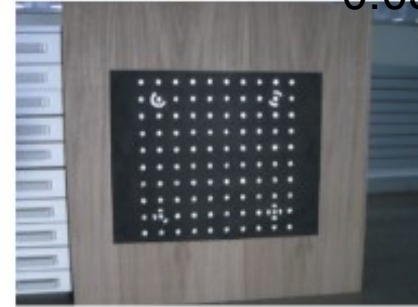
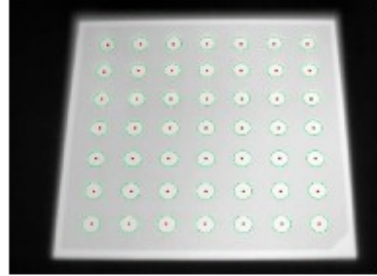
Electric
heating
Resistor

- Heated bulbs on wood/plastic back
- RMSE 0.588

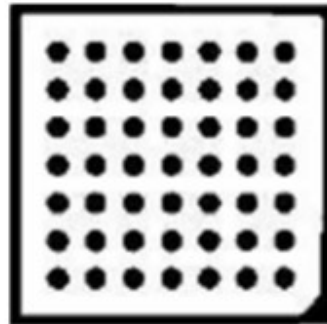
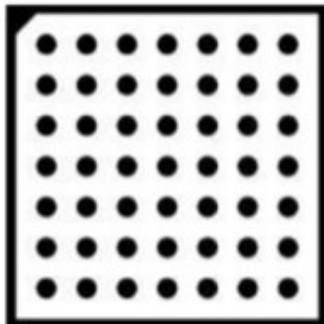
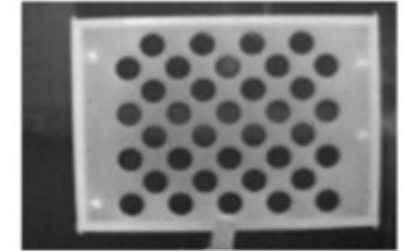


Passive – Circle masks [best]

RMSE
0.052



Cardboard,
RMSE 0.5



Discussion

- It seems that calibration of these thermal cameras is a 'solved' problem
 - RGB cameras tend to have about 0.1 - 0.5-pixel RMSE; which we can already see is met in the literature (i.e. 0.057 RMSE)
- So what are the big issues with most of these systems? Big driver of cost?
 - Metal drives cost -> 24x24 board is \$55
 - Active systems drive cost
- Best active systems already perform worse than the existing passive *cardboard* design(0.5 RMSE), so what to improve on?
 - Maybe Deep Learning or online calibration schemes?